Q1.

Paul had four substances:

citric acid	copper sulphate	indigestion tablet	sugar

He dissolved 1 g of each substance in 20 cm^3 of distilled water.
He used universal indicator to find the pH of each solution.

(a) (i) Sugar solution does **not** change the colour of green universal indicator.

What does this tell you about sugar solution?
Tick the correct box.

It is an acid. ☐ It is an alkali. ☐

It is neutral. ☐ It is sweet. ☐

1 mark

(ii) Suggest the pH of citric acid.

.....

1 mark

(iii) Indigestion tablets neutralise acid in the stomach.

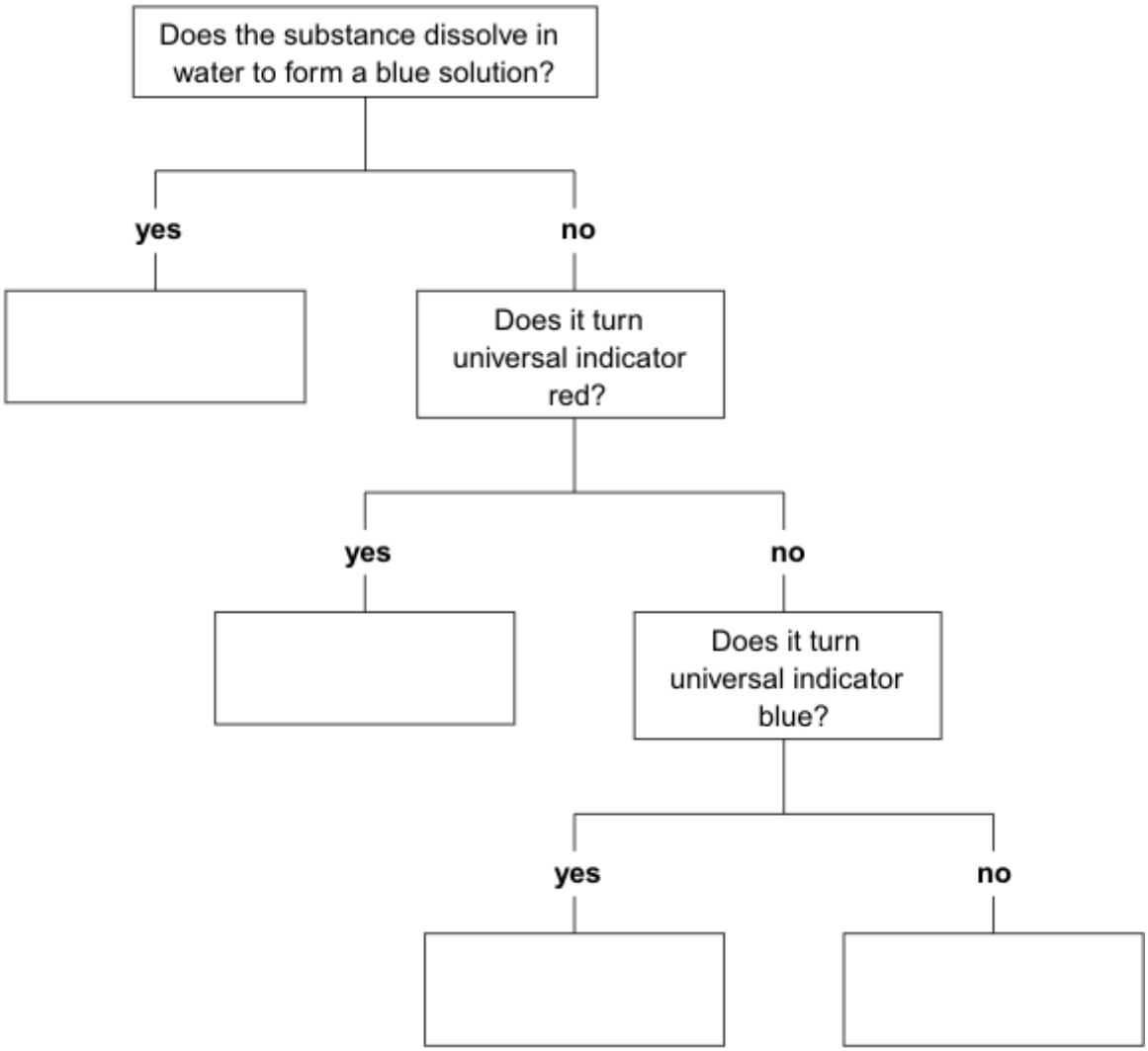
What does this tell you about indigestion tablets?

.....

1 mark

(b) Complete the flow chart below with the names of the substances in the boxes.

citric acid	copper sulphate	indigestion tablet	sugar



3 marks
maximum 6 marks

Q1.

(a) (i) • It is neutral ✓

if more than one box is ticked, award no mark

1 (L5)

(ii) • any number less than 7

1 (L6)

(iii) any **one** from

• they are alkaline **or** an alkali

accept 'a soluble base'
accept 'it has a pH above 7'
specific pH values such as 'its pH is 8.5' are insufficient

• they react with the acid

1 (L5)

(b) • copper sulphate

accept 'copper sulfate'
accept 'the sulphate' for 'copper sulphate'

• citric acid

accept 'acid' for 'citric acid'

• indigestion tablet • sugar

accept 'tablet' for 'indigestion tablet'
*if **all four** answers are correct, award three marks*
*if **two or three** answers are correct, award two marks*
*if **one** answer is correct, award one mark*

3 (L5)

[6]

-